LSTM Training for Sentiment Analysis: Word Influence on Prediction

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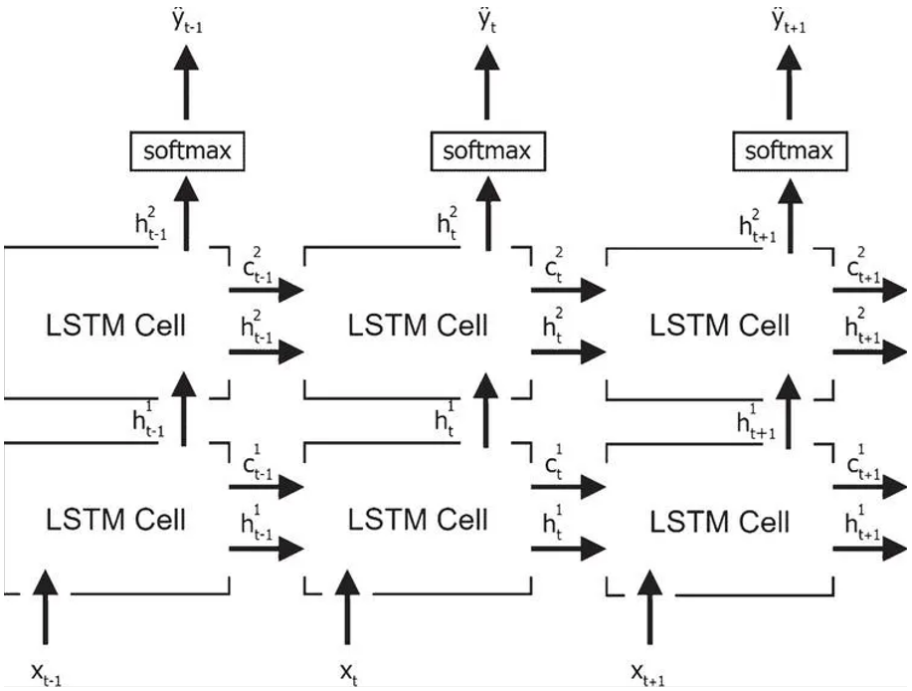
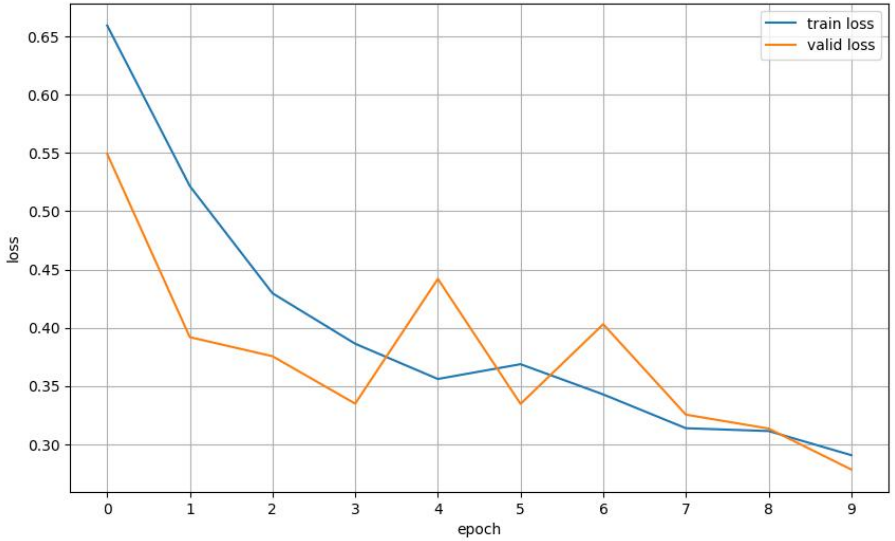
Introduction: RNN Architecture

2-layered LSTM structure





Introduction: Question

Question: **How much does each word contribute to the final prediction?**

- **Hypothesis:** Qualitative adjectives expressing positivity and negativity will contribute the most to the final prediction
- A possible strategy to explore this question: **Gradient x Embedding, time-based analysis, gate analysis.**

Approach: Gradient x Embedding

Example: *"The worst film ever"*

- Word tokenization: $[x_1, \ldots, x_4]$=["The", "worst", "film", "ever"]
- Learned embedding: $x \hookrightarrow \mathbf{e} \in \mathbb{R}^{300}$
- LSTM prediction: $\hat{y}(\mathbf{e}_1, \ldots, \mathbf{e}_4) = 0.17$ (negative review)

Which is the importance of "worst" ($\mathbf{e}_2$) for predicting $\hat{y}$?

Gradient x Embedding: The Idea

Importance

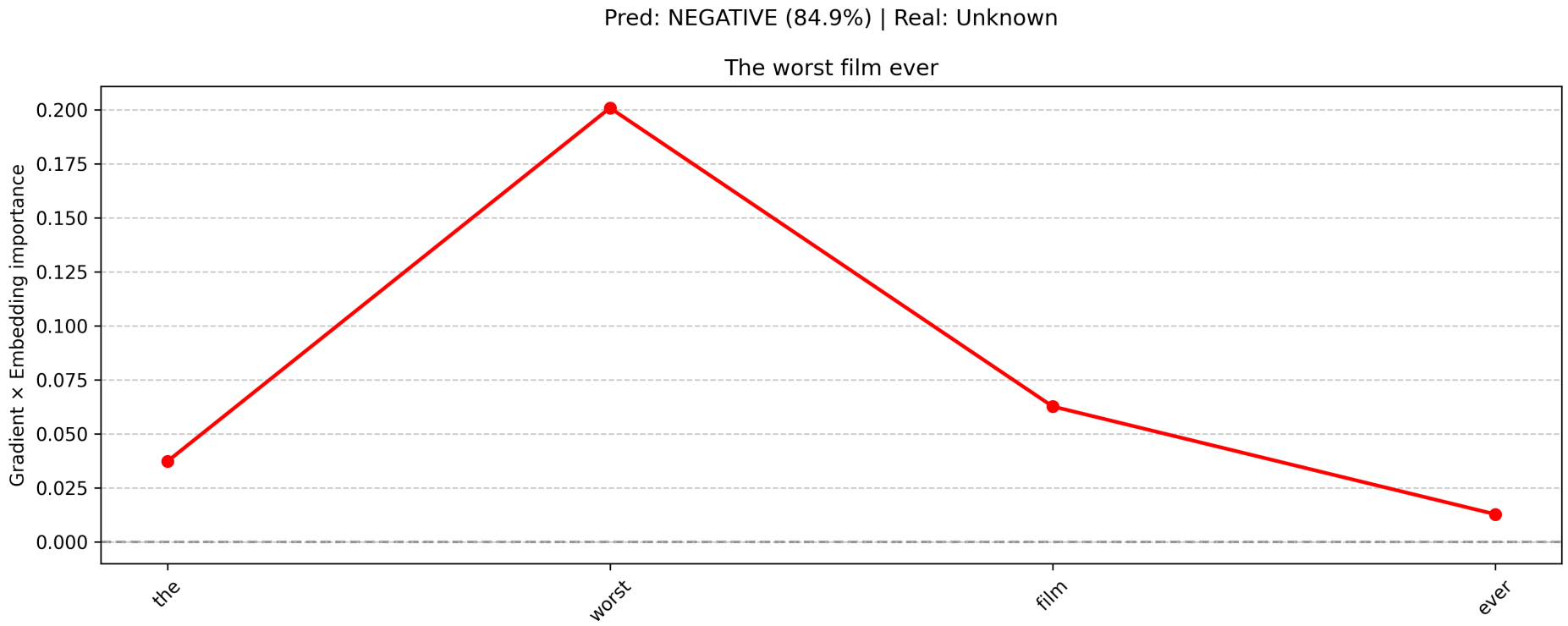
The **Gradient x embedding importance** of the $i-$th word in the review for predicting $\hat{y}$ is

$$\text{Importance}(i) = \underbrace{\mathbf{e}_i}_{\text{magnitude}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial \mathbf{e}_i}}_{\text{sensitivity}}$$

- Sensitivity as the **direction of influence**: *If we slightly change a word's embedding, how much does the prediction change?*
- Sensitivity (and thus importance) depends on the other words in the phrase: **context**
- Sensitivity is computed via **backpropagation** (reverse-mode automatic differentiation)

Gradient x Embedding: Short-Review Example

- ↑ positive: Pushes towards the predicted class
- ↓ negative: Pushes against the predicted class



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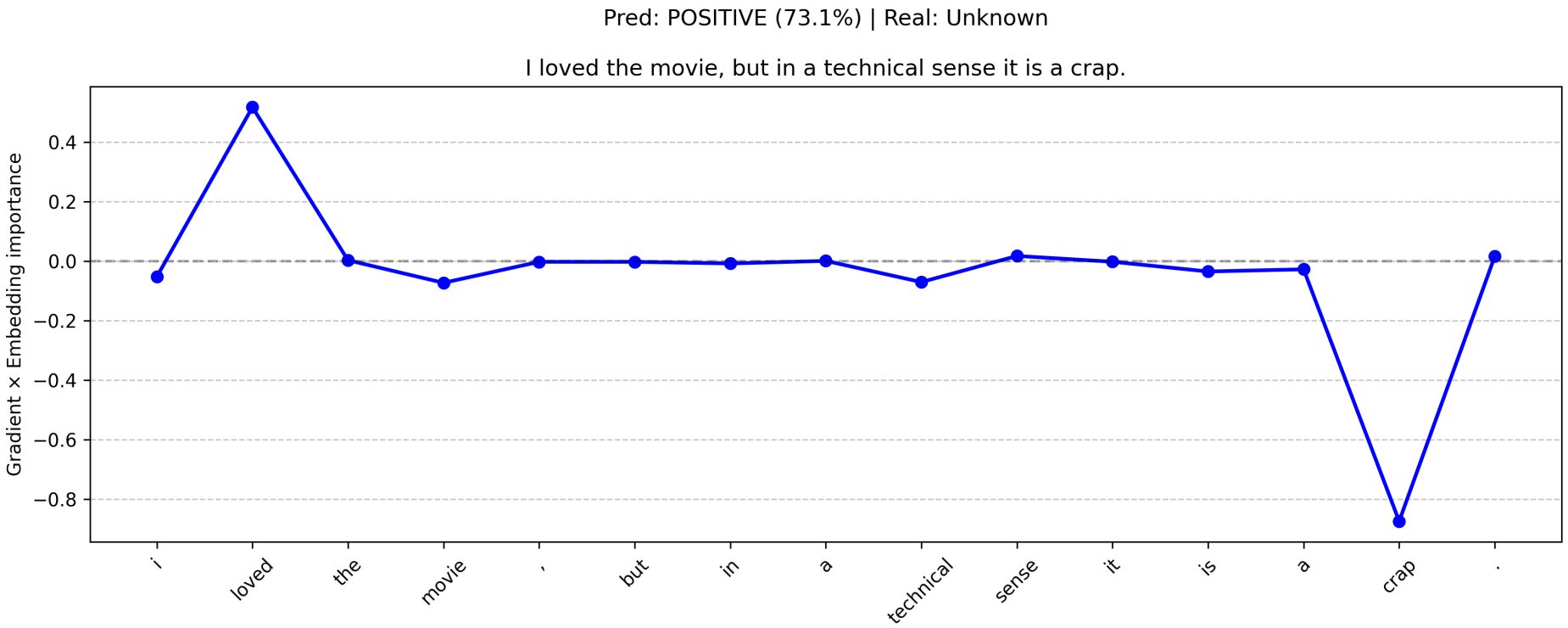

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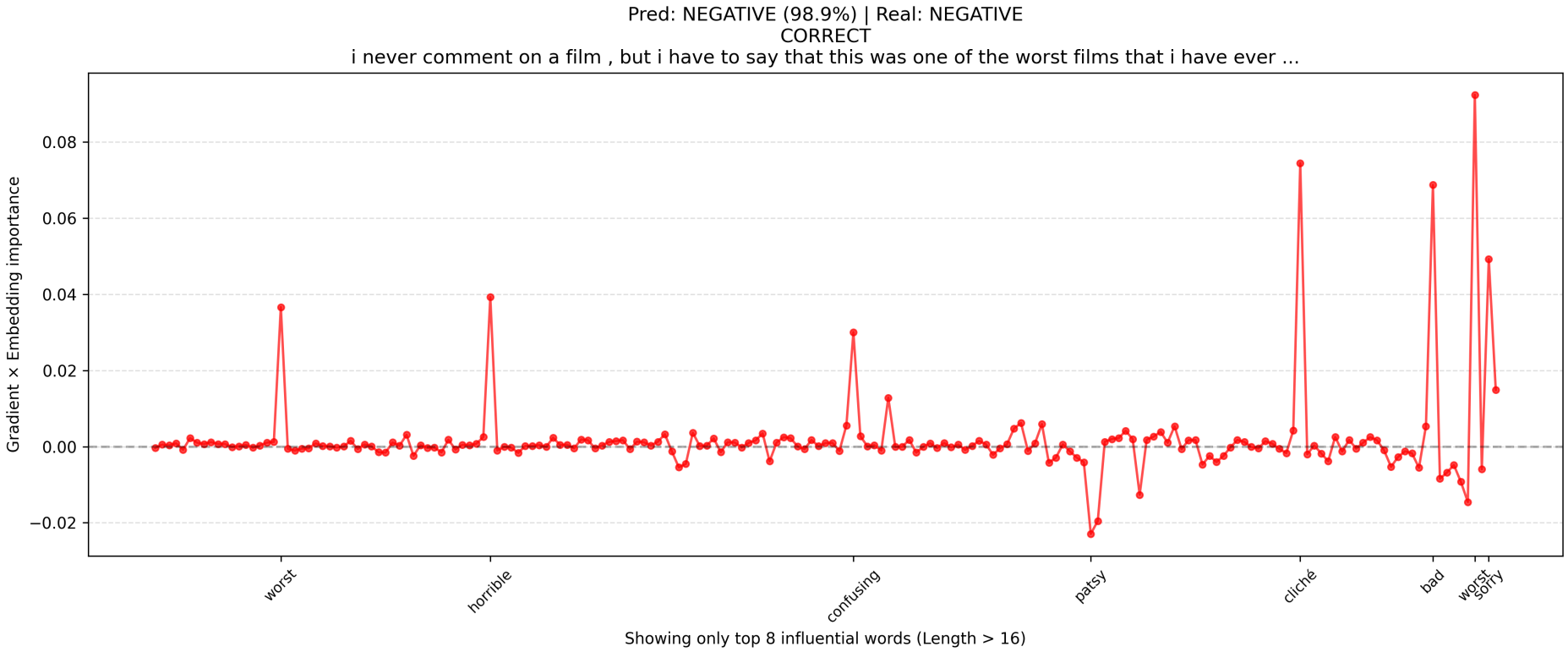

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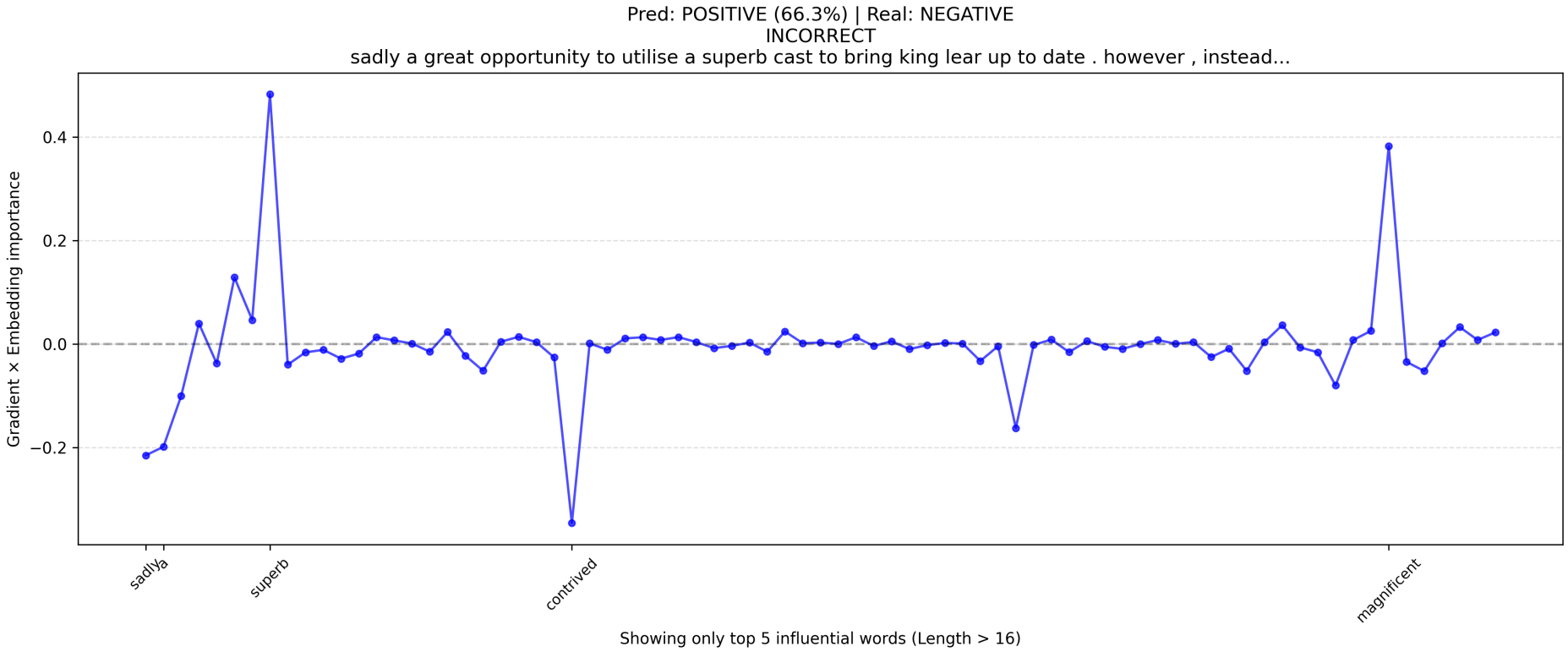
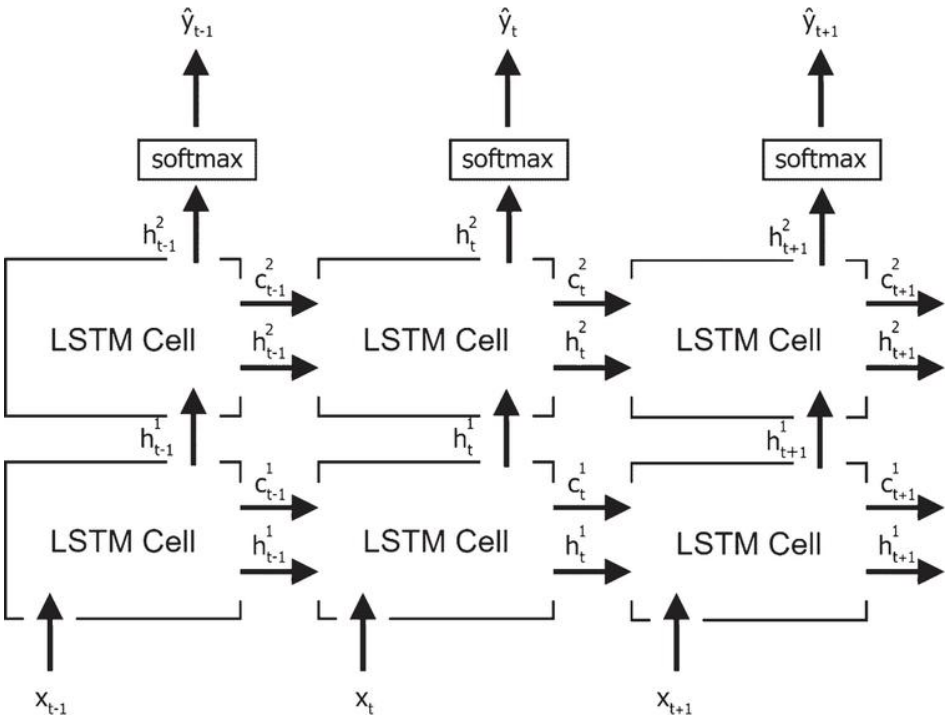


Time-based Analysis in LSTM



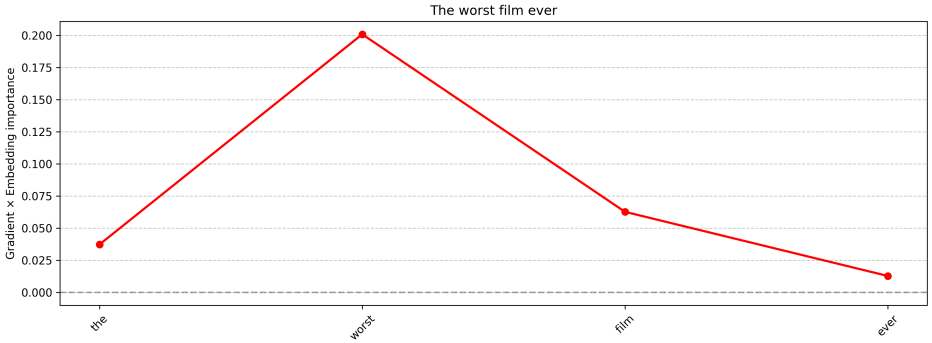
- We evaluate the model's prediction $\hat{y}$ at every sequence step t
- At each time step t, it produces a prediction:

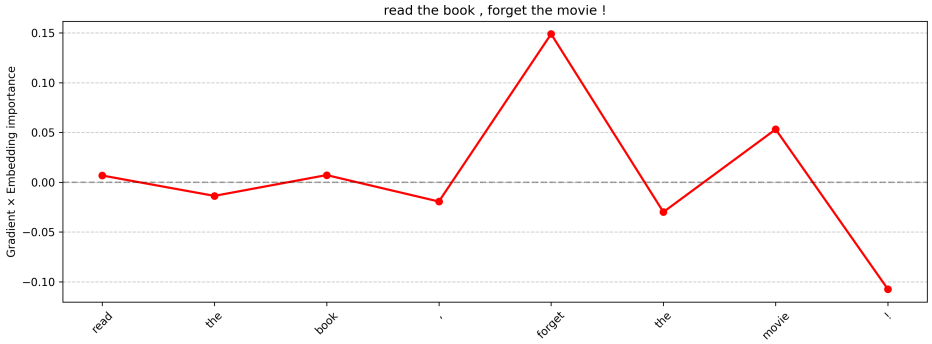
$$\hat{y}_t = \hat{y}(x_1, \ldots, x_t)$$

- We track how $\hat{y}_t$ evolves as new words are added

Short Text Behavior

Gradient x Embedding





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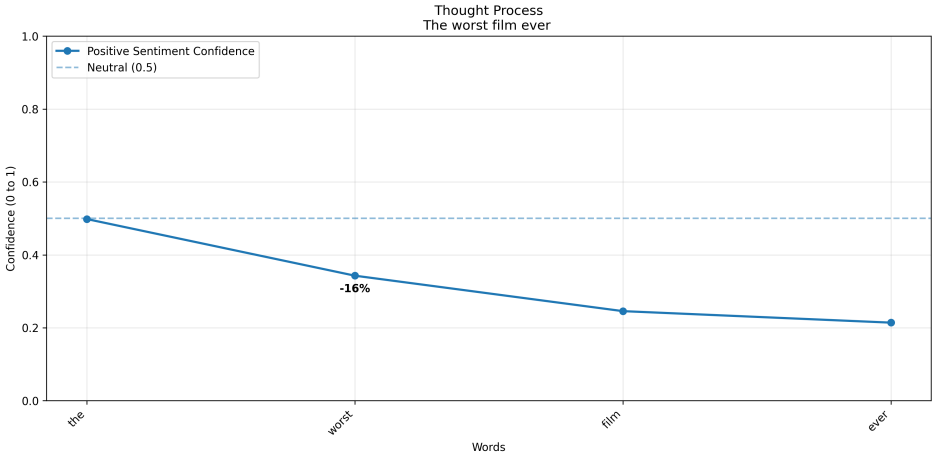

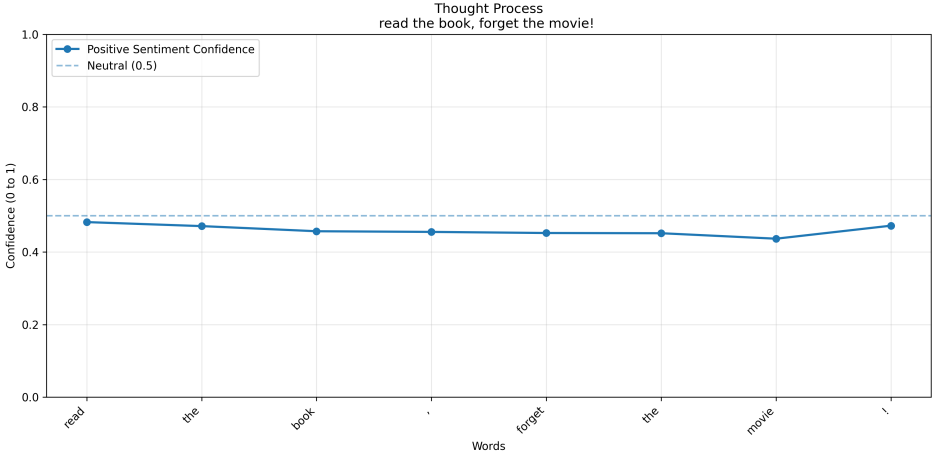

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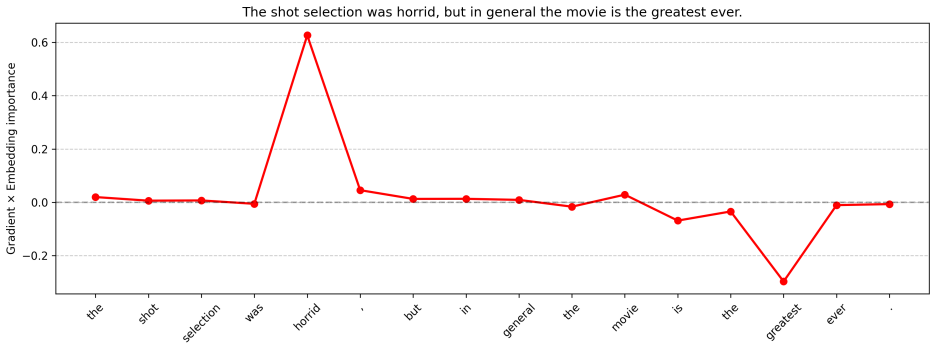

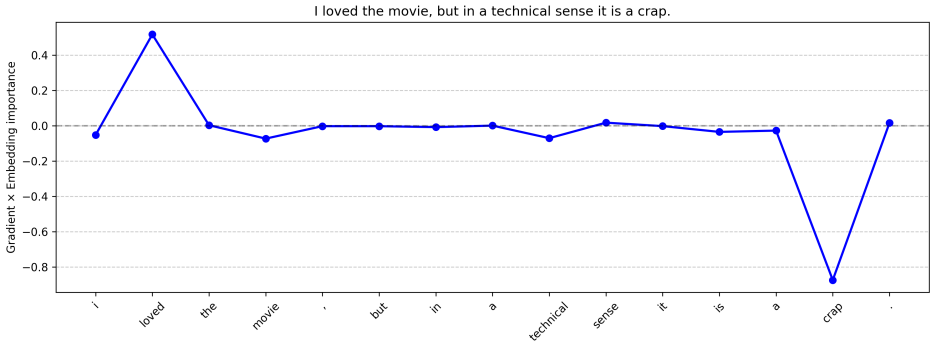

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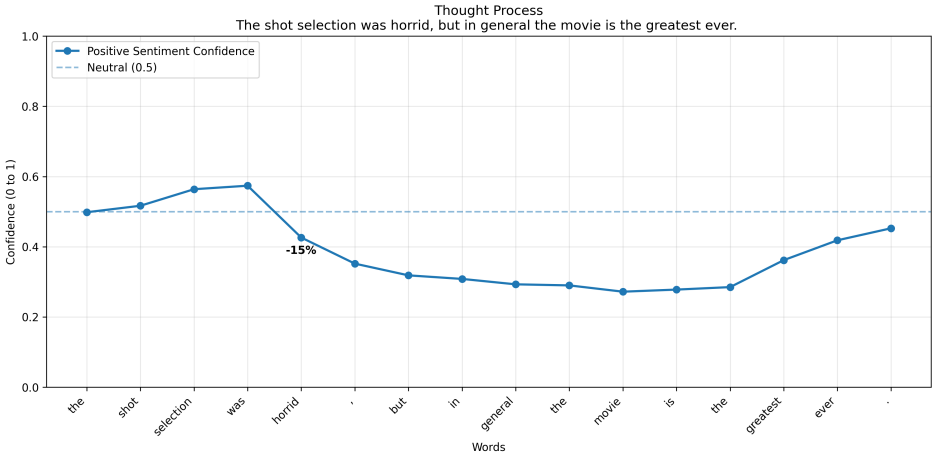
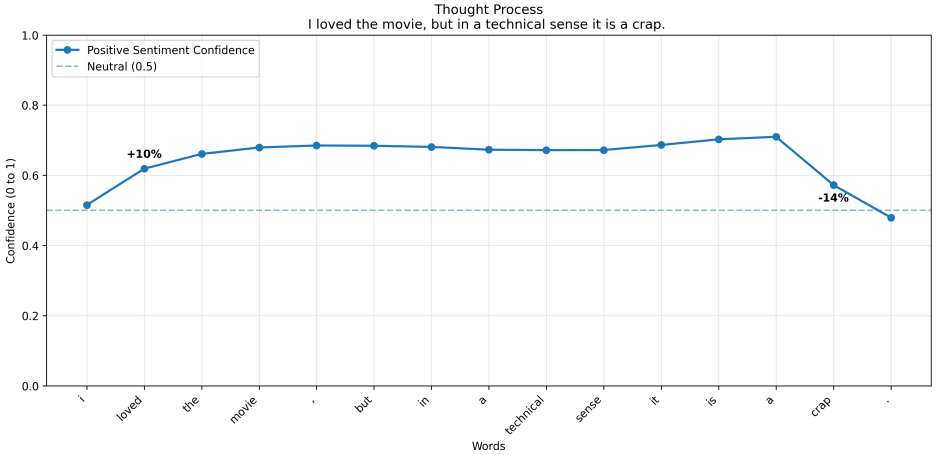
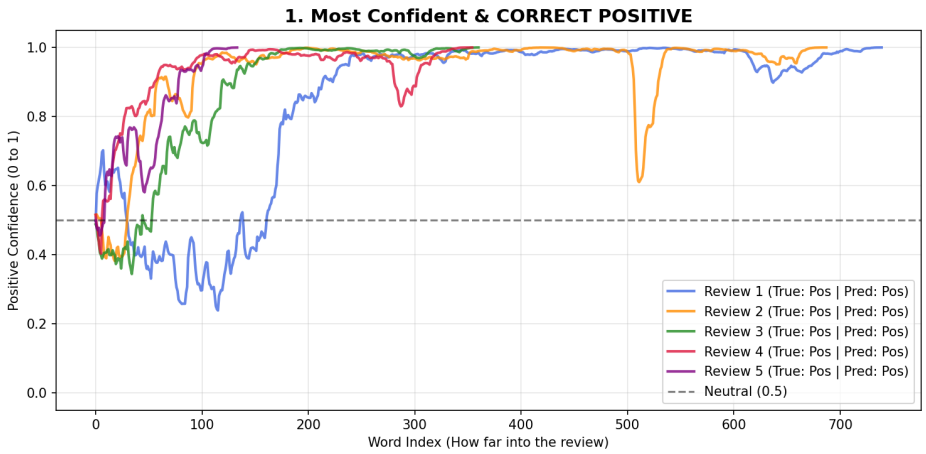


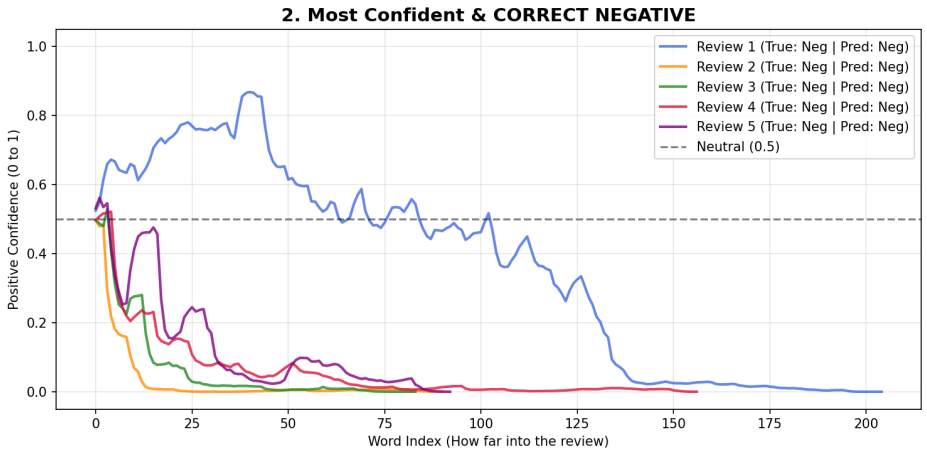


Clear Sentiment Cases

Confident Positive



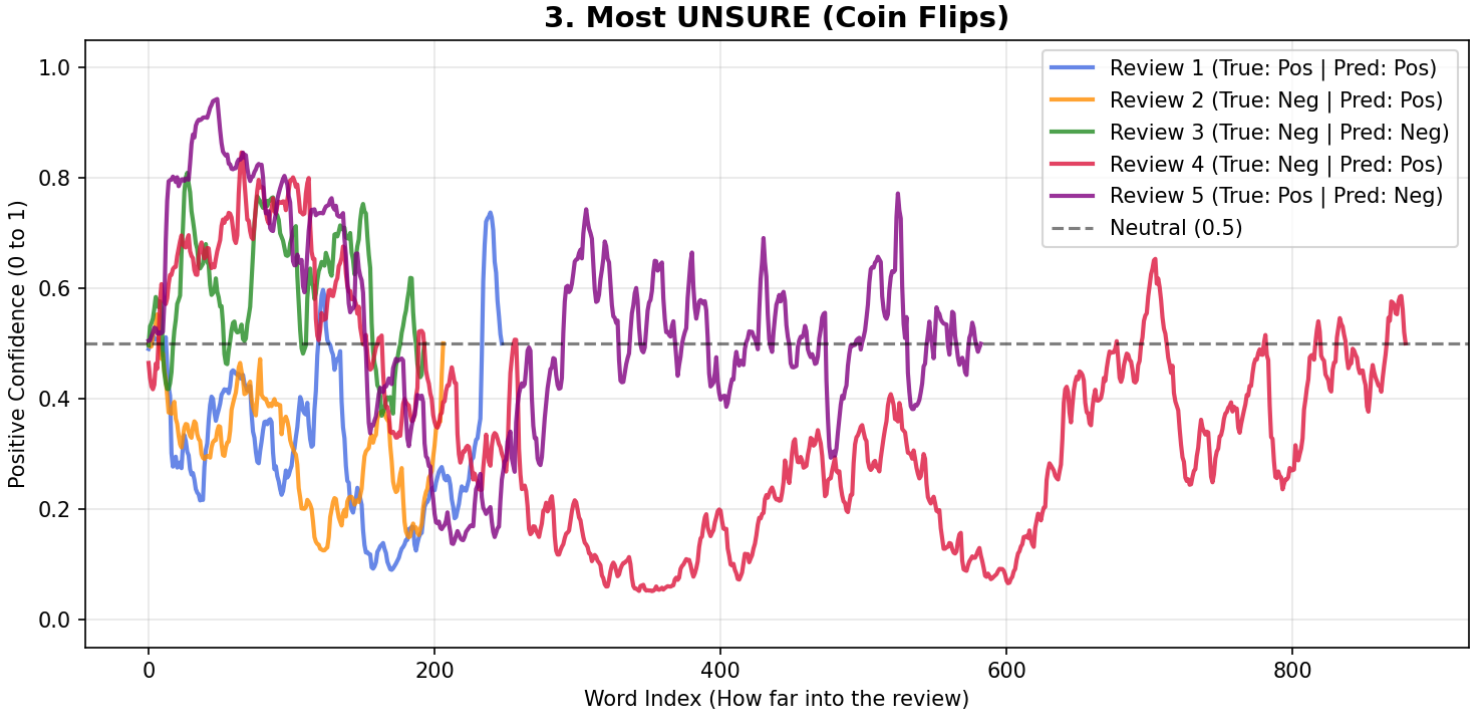
Confident Negative



Review 2 in the confident positive plot showcases an interesting case of the model recovering:

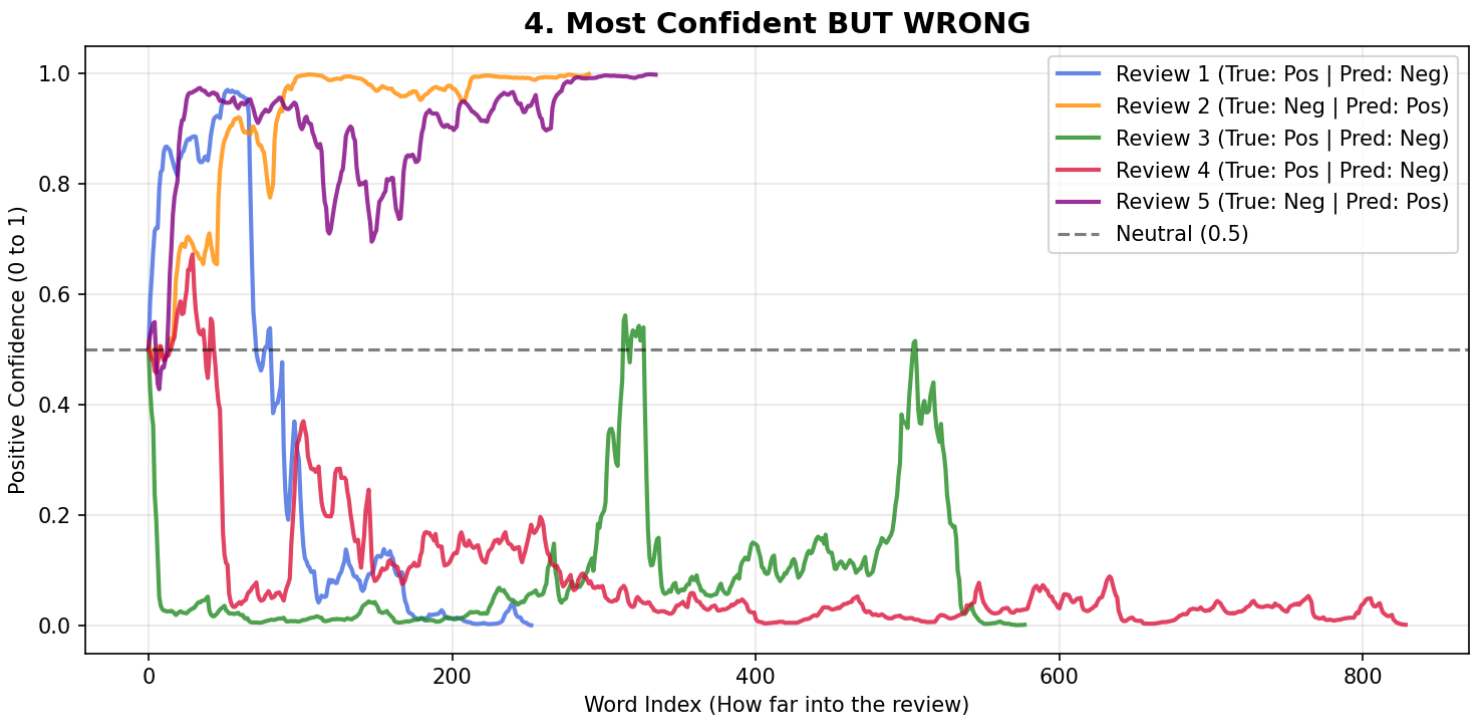
...I can go on, but I won't gush too much more.This film should have received Oscar nominations, certainly one for Best Picture. The performances, without exception, were all wonderful. Ms. Gill's lovely, sultry voice was a surprising epiphany. And Sir Ian McKellen? 'Nuf said. Awesome EIGHTEEN is the reason I slog through over 200 mediocre to utterly **horrendous films** (some in the $150 million plus range) a year, to find that one treasure, that one exquisite, magical, unique and enchanting, perfect...

Uncertain Predictions



The model remains close to 0.5 when sentiment is mixed or ambiguous, reflecting uncertainty in the prediction.

Confident but Wrong



We observe two different scenarios:

1. **Label noise:** clearly positive reviews labeled as negative
2. **Linguistic complexity:** irony or mixed sentiment (e.g., positive words used negatively)

Examples of Confident Errors (Detailed)

1. Likely mislabeled (Review 2)

I found "Strange Fruit" to be an excellent movie... the acting is wonderful, the cinematography and direction excellent, and the script hugely entertaining and thought-provoking...

I hope "Strange Fruit" gets a wider release soon... an above-average thriller... **Highly Recommended: 7/10**

LABELED NEGATIVE
MODEL FINDS POSITIVE

2. Linguistic complexity / irony (Review 1)

...an **entertainingly brainless** piece of **lovably lousy** dreck...

...supremely inane story... silly material... no tension... tacky special effects...

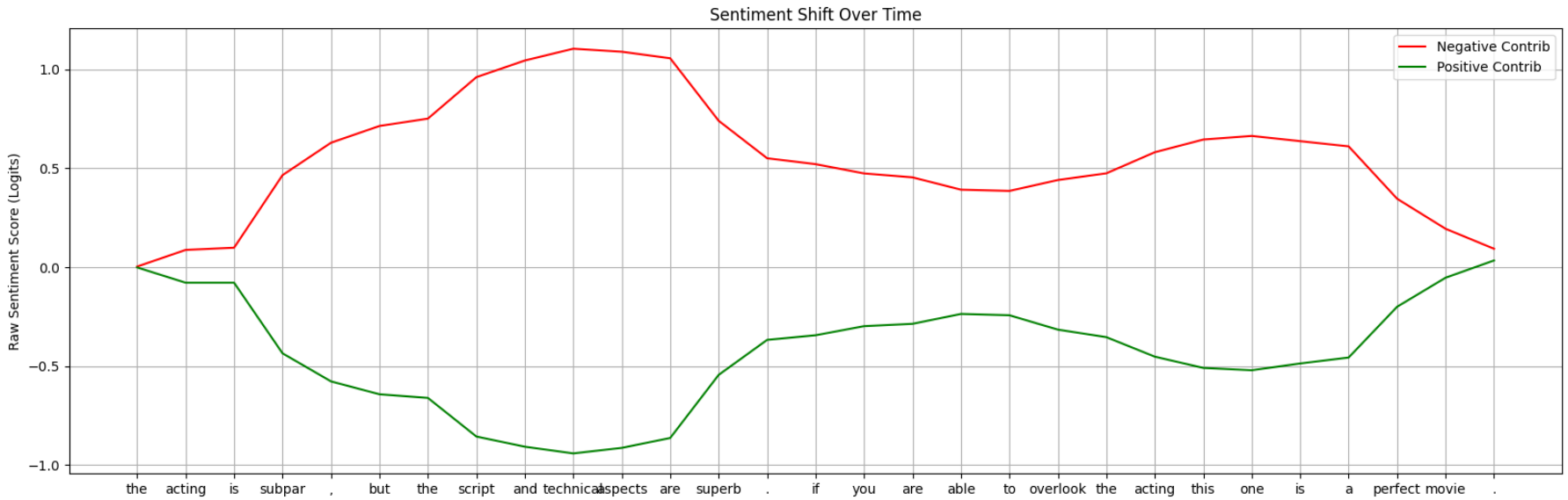
...amusingly lowbrow... delicious cheesiness...

LABELED POSITIVE
MODEL FINDS NEGATIVE

LSTM: Gate analysis

Idea: Even though certain words carry considerable connotations, context is still important to understand a review. From this, we aim to analyze how an LSTM interprets and or uses context for its interpretation of a review.

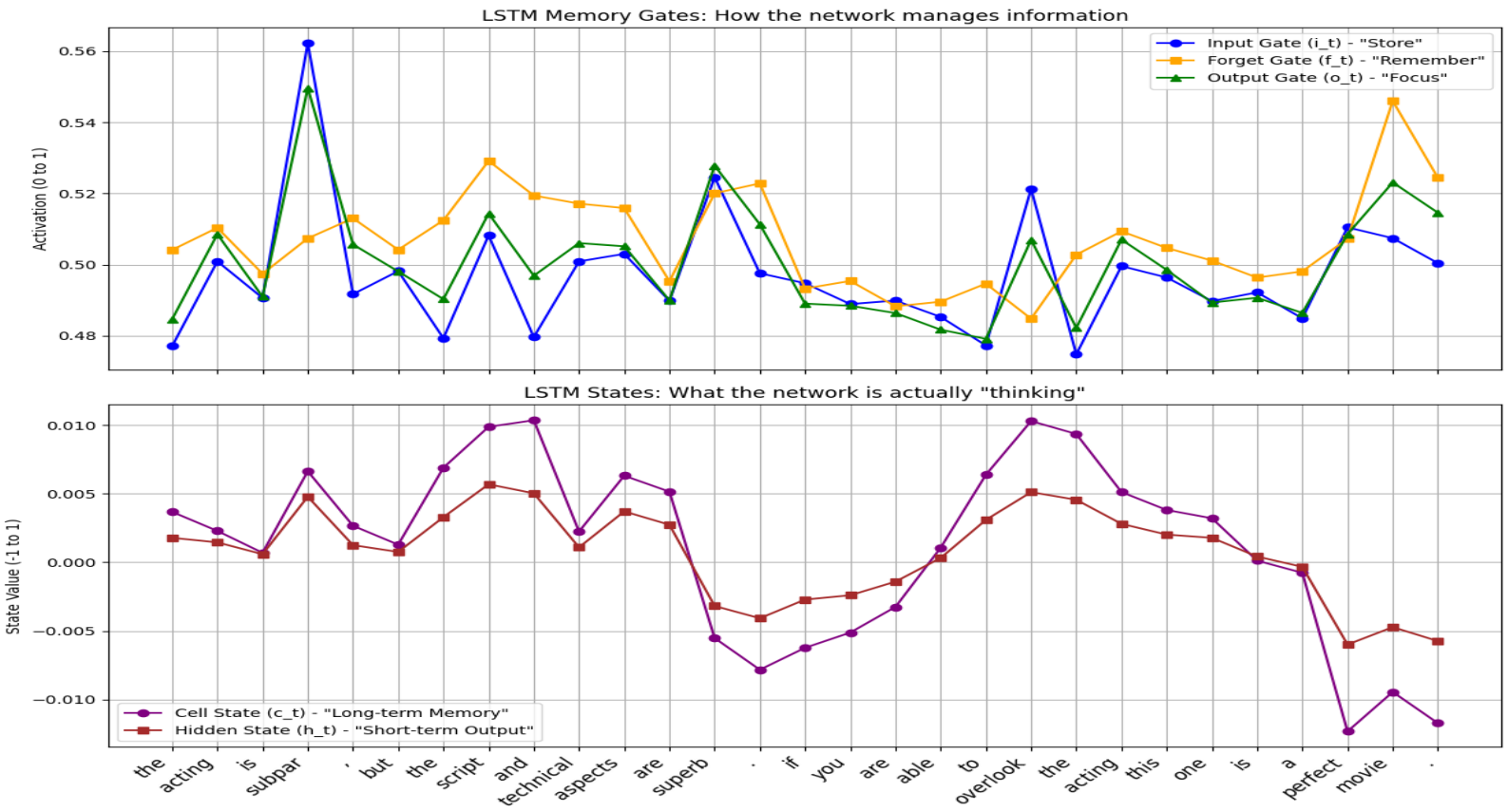
LSTM Analysis: The Conflicted Review



The Tug-of-War: A Neutral Convergence

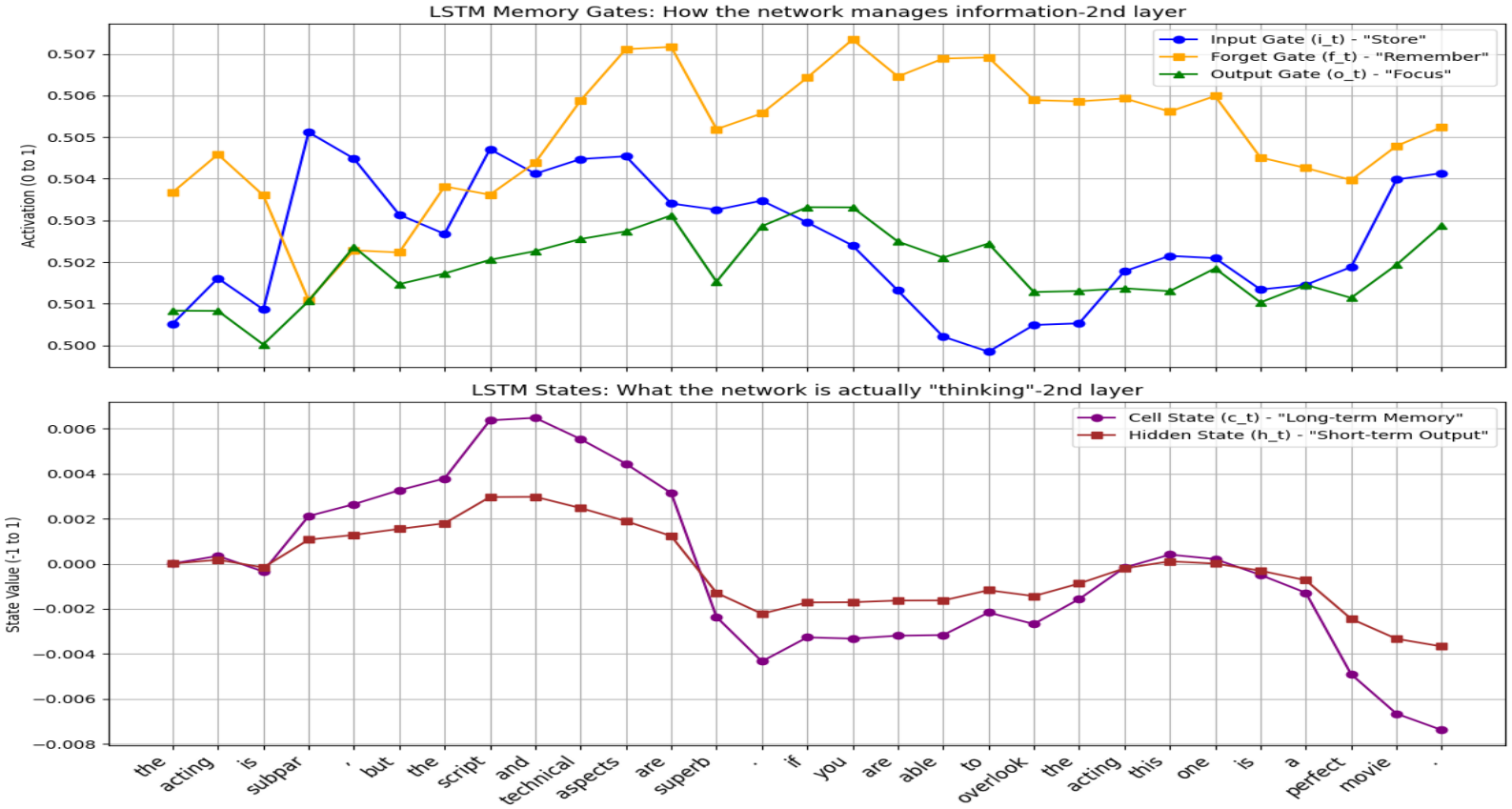
- **The Negative Start:** "subpar" pushes the negative logit (red) high.
- **The Reversal:** "superb" begins dragging the prediction back.
- **The Conflict:** "overlook the acting" prevents a full positive takeover.
- **The Convergence:** Logits meet near zero, proving the model balances opposing context rather than just reading the final words.

LSTM Analysis: Layer 1



- **High Volatility:** Gates react violently to local grammar and individual words.
- **No Big Picture:** Short-term memory focuses purely on immediate syntax, unaware of the broader conflict in the review.

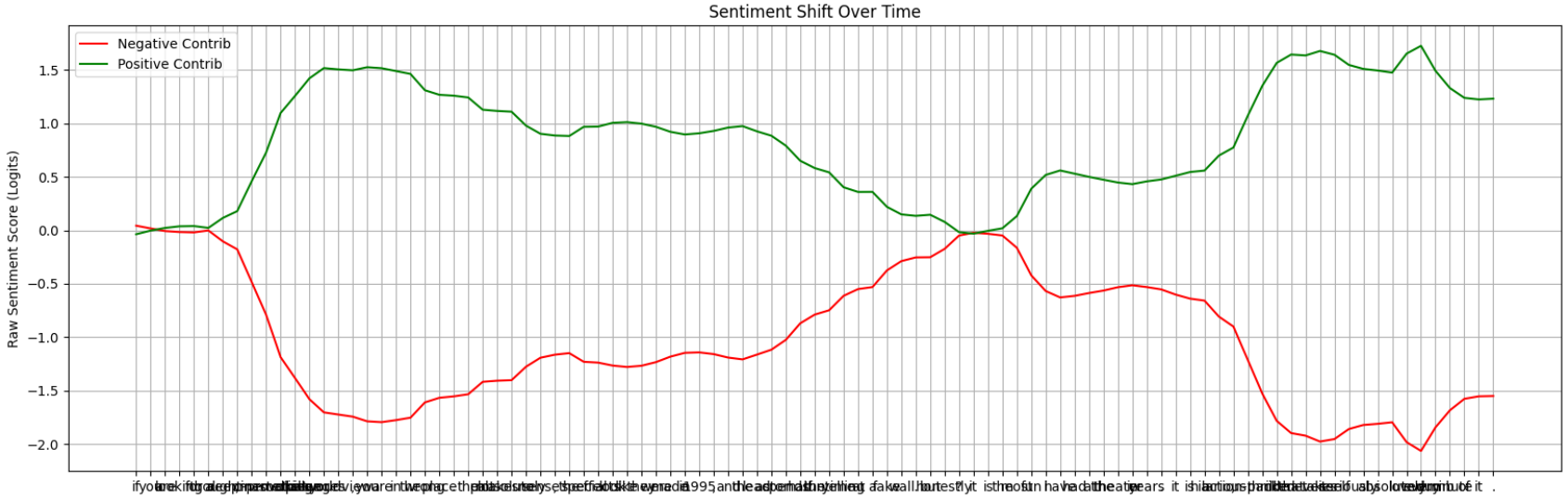
LSTM Analysis: Layer 2



- **The Pendulum Swing:** Cell State (c_t) drifts up (negative), plunges (positive), and climbs again, perfectly mapping the conflicting clauses.
- **The Power of Memory:** The Forget Gate (f_t) stays high throughout the second half, refusing to "forget" the bad acting even when reading "perfect movie".

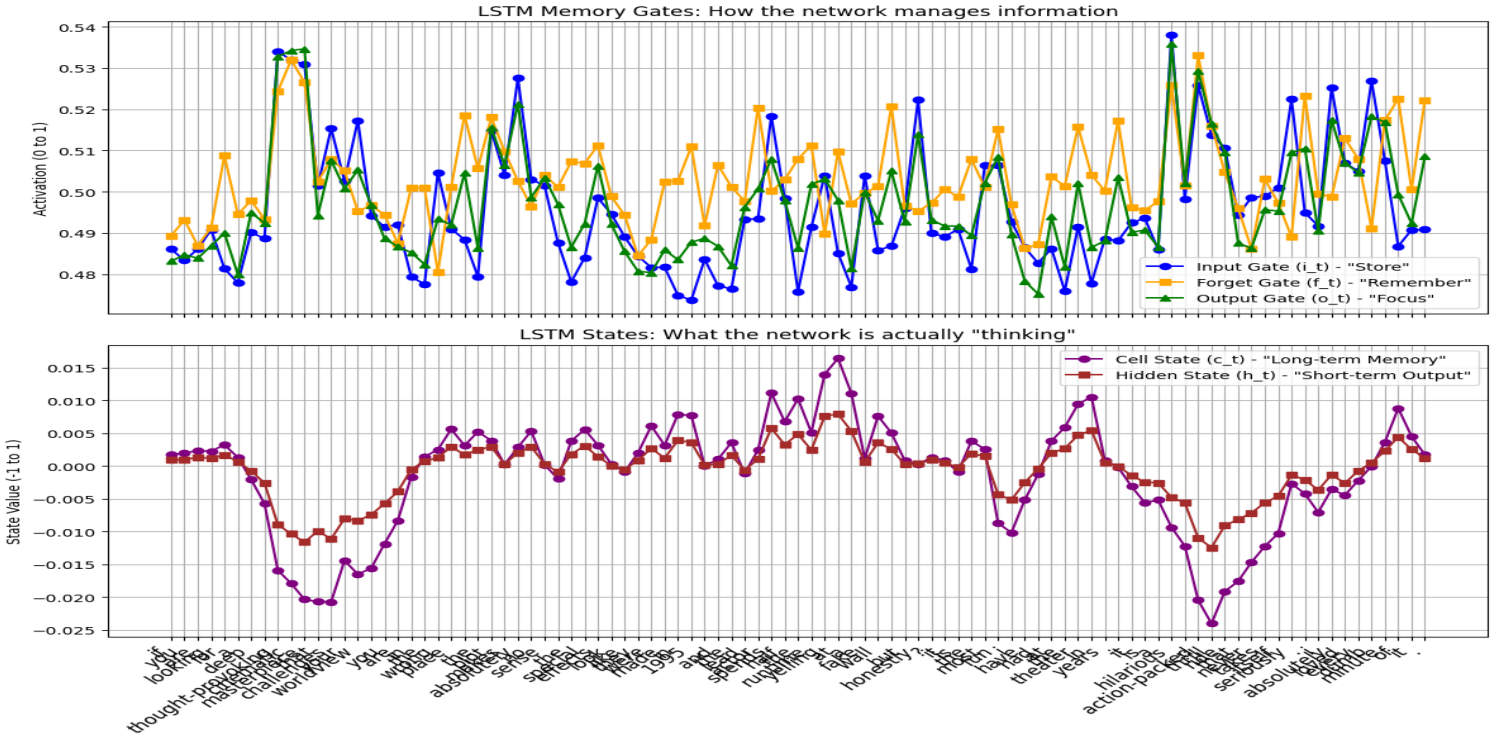
LSTM Analysis: The "Sarcastic Pivot"

"If you are looking for a deep, thought-provoking cinematic masterpiece that challenges your worldview, you are in the wrong place. The plot makes absolutely no sense, the special effects look like they were made in 1995, and the lead actor spends half the runtime yelling at a fake wall. But honestly? It is the most fun I have had at the theater in years. It is a hilarious, action-packed thrill ride that never takes itself seriously. I absolutely loved every dumb minute of it."



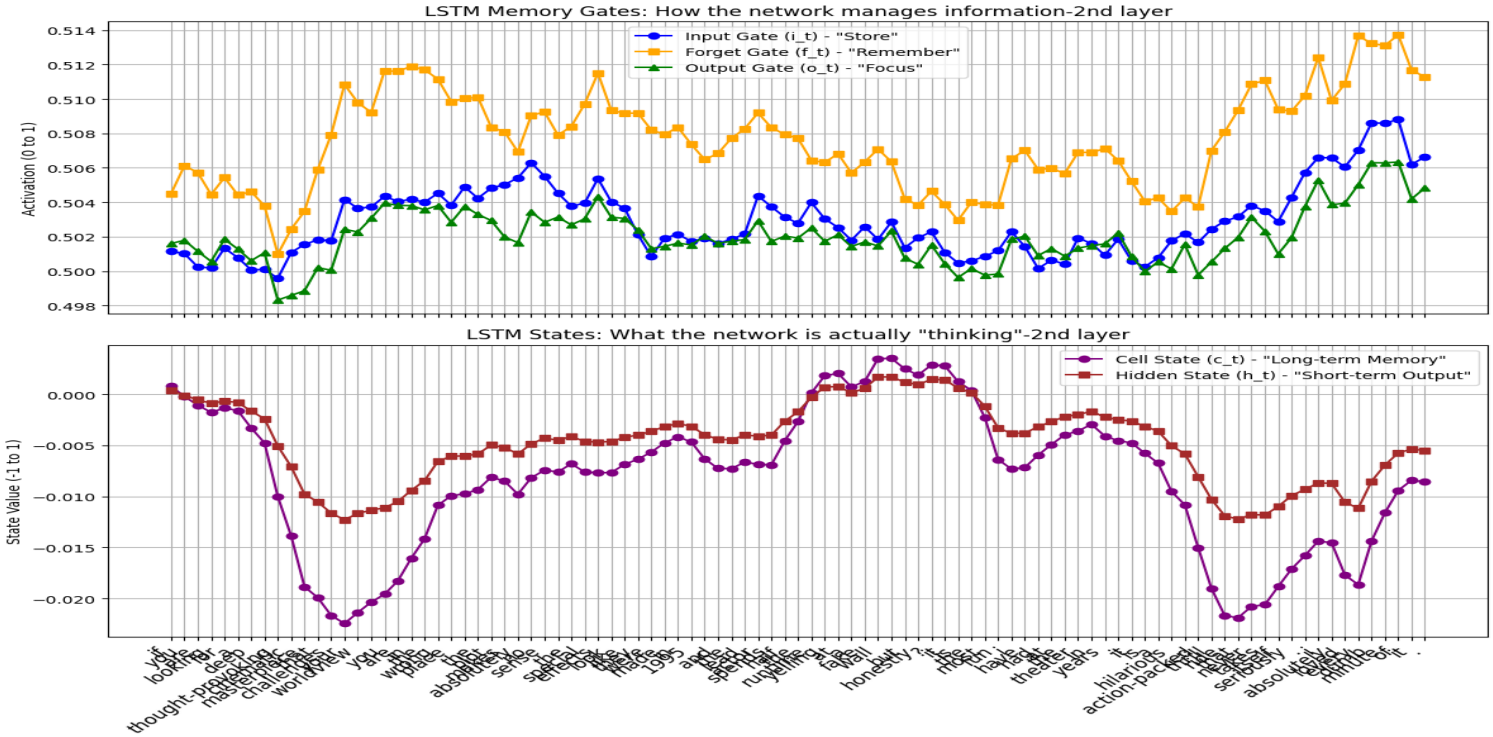
- **The Bait:** "Cinematic masterpiece" fools the model into a positive prediction.
- **The Trench:** 30+ words of negative complaints drag the sentiment down.
- **The Reset:** "But honestly?" neutralizes the score, preparing for the climax.

LSTM Analysis: Layer 1



- **High Volatility:** Notice the extreme thrashing of the cell state (c_t) and gates.
- **Reflexive Parsing:** The Input (i_t) and Output (o_t) gates spike aggressively on almost every single word.
- **No Context:** This layer has zero stamina. It processes raw grammar and individual words without understanding the overarching narrative.

LSTM Analysis: Layer 2



- **The Patient Forget Gate:** The orange line (f_t) stays remarkably high during the negative trench, deliberately keeping the context window open.
- **The Narrative Arc:** The Cell State (c_t) traces a smooth wave-drifting into the negative zone, turning sharply at "But honestly", and diving back to positive.
- **Conclusion:** Layer 1 reads the *words*, but Layer 2 reads the *story*.

Conclusions: Words vs. Context

How does the LSTM make its final prediction?

- **Words Provide the Signal:** As seen in the Gradient $\times$ Embedding analysis, specific qualitative adjectives (e.g., "worst", "superb") act as massive triggers, causing sharp, immediate shifts in the model's confidence.
- **Context Dictates the Verdict:** However, the LSTM does not simply add up word scores. It actively manages long-term context to handle sarcastic pivots and conflicting clauses.
- **The Mechanism of Understanding:** By utilizing a hierarchical 2-layer structure, the network delegates raw syntax parsing to the bottom layer. This frees the top layer's Forget Gate (f_t) and Cell State (c_t) to patiently hold contradictory information in its memory until the narrative resolves.

Final Takeaway: *Individual words might steer the network, but the LSTM's internal memory ensures that the overarching story drives the final prediction.*